Game Theory and Social Choice

Lecture 14: Akerlof and the Market for Lemons

Brian Weatherson

2026-10-15

The Used-Car Discount

Recap

Spence asks what the informed side can do about asymmetric information. And he suggests a natural answer: pay for a signal the other side won't/can't pay for. (He says won't; I think can't is much more plausible given the numbers.)

Akerlof considers the case where the more informed side can opt out of the market.

We're working backwards here; Akerlof's paper came out in 1970, and was one of the influences on Spence's paper. (In his Nobel lecture, Spence mentions his thesis advisor, Kenneth Arrow, encouraging him to read Akerlof's paper.)

A Fact That Needed Explaining

For most of the twentieth century the used car market had a peculiar feature.

Cars lose value with age and mileage, and for most of a car's life that decline is reasonably smooth.

But there was a cliff at the start. Drive a new car off the lot and it lost a large chunk of its value at once, at least 10%, perhaps 20%. Everybody knew this, but there were no really good theories as to why, and a bunch of fairly bad theories.

Akerlof's Idea

Most cars are good. Some are duds, which in the language of the period are **lemons**.

A test drive, i.e., driving the car around the block a couple of times, doesn't reveal that it's a lemon. But a few weeks at home is enough to know.

So the seller knows and the buyer does not, and no amount of kicking the tires or doing reasonable test drives fixes that.

A Very Simple Model

Restrict attention to people who have a car and would, other things equal, like to sell it at the market price.

Assume half the cars are lemons.

We're making two very silly assumptions here, namely that half are lemons and that they are evenly distributed over sellers and non-sellers, but it keeps the arithmetic clean. We'll make a more plausible model in a second.

A Very Simple Model

Table 1: Values in the extreme version

	To seller	**To buyer**
Good car	100	120
Lemon	40	60

If Everyone Could See

Every car is worth more to a buyer than to its seller. A good car is worth 20 more, and so is a lemon.

So with full information every car changes hands, every sale produces a surplus of 20, and the market does what markets are supposed to do. Surplus value everywhere, everyone is happy, aren't markets great, all praise markets, etc.

What a Buyer Will Pay

Now hide the quality. A buyer who thinks the odds are even values a random car at

$$\frac{120 + 60}{2} = 90$$

So she will pay up to 90, and no more.

Who Sells at 90

A good car is worth 100 to the person who owns it.

At 90, no owner of a good car sells.

So every car actually on the market is a lemon.

The Revision

A buyer who works this out revises. A car on offer is worth 60 to her, since it is a lemon.

Buyers now offer at most 60. Lemon owners value their cars at 40, so they sell happily.

Where It Ends Up

The market clears, in the sense that the cars offered get sold. Only lemons trade, at somewhere between 40 and 60.

Good cars never change hands.

Every one of those sales would have produced a surplus of 20, and none of them happens.

Nobody messed up, but we left some surplus value on the table. That’s bad.

Who Actually Sells

What We Have Not Explained

The fact we started with was a **price**: a car loses something like 20% of its value the moment it is driven off the lot.

What the model has delivered so far is that good used cars do not sell at all. That's not what the real world situation was like. (Maybe it was what the real world model for cars a few weeks old was, not the general case.)

To get a price for a good used car, we need owners of good cars who will take less than 100.

A Second Split

Sellers differ in more than one way.

1. How good their car is;
2. How motivated they are to sell.

Again, no calculus here, so we're making both of these binary divisions. That gives us four kinds of seller.

And We Can Drop the Absurd Part

While we are here, we no longer need half the cars to be lemons. Take a hundred cars, of which 90 are good and 10 are lemons.

Of the 90 good cars, 80 belong to people who are happy with them and are not selling at any price a buyer would offer.

That leaves 20 cars whose owners might sell, and those divide four ways.

Four Kinds of Seller

Buyers are unchanged from before: they'll pay 120 for a good car, 60 for a lemon.

Table 2: The twenty cars that might be sold

Seller	Will accept	Cars
Good, relaxed	100	5
Good, pressed	70	5
Lemon, relaxed	40	5
Lemon, pressed	20	5

Where the Even Split Came From

One car in ten is a lemon. One car in two *on the market* is a lemon.

You don’t need to have a catastrophic quality control for that to happen.

What you do need is that the proportion of badly made cars is similar to the proportion of badly made decisions to buy cars; i.e., decisions made by people who really need to sell soon after.

So there can be lots of lemons on the market not from raw factory quality, but as a selection effect. (This might be like the catastrophe that Zillow encountered.)

Finding the Price

A price works when buyers break even on it: what they get on average is worth at least what they pay.

If the price is 100, all twenty sell, so the pool is half lemons and worth 90. Buyers lose 10 a car, so 100 is too high an estimate of the price.

What if we think the price is 80, Now the relaxed owner of a good car keeps her car, since 80 is below her 100. The other three groups sell.

Eighty Holds

At 80 the pool is fifteen cars: five pressed good ones and ten lemons. One third of it is good.

$$\frac{1}{3} \times 120 + \frac{2}{3} \times 60 = 80$$

Buyers break even exactly, so nobody has a reason to bid differently. That works as an equilibrium.

Now There Is a Discount

A good car is worth 120 to a buyer, and on the used market it fetches 80.

That is a third off a good car, and it kicks in the moment it becomes a used car, which is the phenomenon we started with.

The people who accept it are the people who have to sell.

Who Leaves

The relaxed owner of a good car levaes the market. Her car is worth 100 to her and the market is offering 80.

That costs 100 of surplus: five cars at 20 each. Every other sale still happens.

The market shrinks and tilts, rather than collapsing.

A Better Fit

The two-type model said good used cars never change hands, which is false.

This one says the used-car market exists, runs at a discount, and over-represents cars sold under pressure. That is roughly what we see.

And a car being up for sale is still evidence that it’s a lemon. The discount is what buyers demand in order to overcome that evidence.

Signalling Again

Offering It for Sale Is a Move

Look at what the seller’s decision has become.

Putting a car on the market is a signal, and what it signals is that you are the unfortunate owner of a lemon.

Nobody intends to send it. Too bad, put your car up for sale and you’re sending just that signal.

Predicting the Fading

Two Conditions

The model needs variation in manufacturing quality large enough to make lemons a serious worry.

And it needs a real information gap between buyer and seller.

Both Have Weakened

Quality control is much better than it was. The cars coming off a line now resemble each other closely.

Vehicle history services and cheap independent inspections have narrowed the information gap, and manufacturer certification covers a fair bit of what is left.

How Would We Check?

The drop when you drive off the lot won't tell us. It mixes adverse selection with dealer margin, sales tax, the warranty, and plain depreciation.

Here's a better idea: price the signal. If selling is still a signal of bad quality, a credible guarantee of quality should be worth a lot.

Pricing the Signal

I couldn't find any *really* up to date studies on this, but I found one from the late 2010s. (We want to go there anyway, because the COVID-era used car market was really weird.)

Matched on trim, body style, drive type and mileage, across a million used cars, certified pre-owned sells for about **3.6% more** than non-certified, or around $800.

And that is a ceiling, not an estimate. Certification bundles a warranty, which is worth money even if buyers could see everything.

So the most that resolving the asymmetry can be worth here is a few hundred dollars on a $20,000 car.

The lowest spread is for Honda, who I guess make reliable cars. But I'm not sure how much to read into that, since the spread was pretty high for Lexus, and I would have thought they had a pretty good production line as well.

Why That Matters

Explaining a striking phenomenon is one thing.

Saying in advance which changes would make it go away, and then watching those changes arrive and the phenomenon fade, is a much stronger result. Not many models in economics manage it.

Economists Disliked It

When the paper first appeared, it was not popular. Economists took it badly. Abandoning perfect information was seen to be a really big step, one that would change how one could possibly do economics.

This isn’t because the agents are internally incoherent.

They have sensible priors, they update by conditionalisation, and then they maximise.

By the standards a philosopher would use, everybody in the model is perfectly rational.

Ignorance and Irrationality

What is true is that the agents are not fully informed.

Economists have since mostly come round to the philosophers' way of putting it. Being ignorant is unfortunate, and it is not a failure of rationality.

The sociology of how the discipline reacted would make a good topic for a different course.

Next Time

That is the game theory half of the course.

After the break we start on social choice, and the question changes from what individuals should do to what a group's preferences even are.